

HARIHARAN NARLAKANTI

Software Engineer | Backend & AI Engineer

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SUMMARY

Computer Science graduate specializing in Artificial Intelligence & Machine Learning with an **8.89/10 CGPA**, focused on software, backend, and AI engineering. Built and deployed AI-powered applications using Python, FastAPI, REST APIs, Hugging Face Transformers, semantic search, RAG, Docker, and GitHub Actions. Hands-on experience in backend API development, document processing, embedding-based retrieval, NLP evaluation, automated testing, and cloud deployment.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

B.Tech in Computer Science & Engineering (AI & ML)

CGPA: 8.89/10

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Backend: FastAPI, REST APIs, Pydantic, JWT Authentication, Node.js, Express.js

AI/ML: NLP, Hugging Face Transformers, PyTorch, SentenceTransformers, RAG, Embeddings, QLoRA, PEFT

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB

Testing & DevOps: pytest, Git, GitHub Actions, CI/CD, Docker, Linux

Deployment: Hugging Face Spaces, Vercel

EXPERIENCE

Virtual Internship – Full Stack Developer (MERN Stack), SmartInternz

May 2025 – Jul 2025

- Built **SocialeX**, a MERN social networking application using React, Node.js, Express.js, and MongoDB with authenticated REST APIs and JWT-based access control.
- Implemented real-time one-to-one messaging with **Socket.IO** and enforced server-side authorization for user profiles, posts, and protected resources.

PROJECTS

Saaram – AI-Powered Telugu News Summarization & Speech Platform

GitHub | Live Demo

- Built and deployed a full-stack Telugu news summarization and speech platform using **Python, FastAPI, React/Vite, Hugging Face Transformers, and Docker**.
- Developed a hybrid NLP pipeline combining morphology-aware **TF-IDF**, pretrained **mT5**, and parameter-efficient **QLoRA** fine-tuning for resource-constrained inference.
- Evaluated **5 configurations on 1,302 XL-Sum Telugu samples**; fine-tuned mT5 achieved **0.1639 ROUGE-1, 0.0496 ROUGE-2, and 0.1452 ROUGE-L**.
- Improved character n-gram TF-IDF **ROUGE-2 by 26.6%** over the word-level baseline and implemented adaptive routing with a resource-aware TF-IDF fallback.
- Automated backend deployment to Hugging Face Spaces using **Docker, GitHub Actions, and OIDC-based Trusted Publishers**.

ResearchCompass – AI Research Paper Analyzer

GitHub | Live Demo

- Built a document intelligence platform using **Next.js, FastAPI, PyMuPDF, SentenceTransformers, ChromaDB, and Groq** for academic paper analysis and semantic retrieval.
- Designed modular REST API workflows for PDF ingestion, section-aware chunking, embedding generation, vector retrieval, and structured LLM inference using **Pydantic**.
- Implemented targeted multi-query **RAG**, cross-paper comparison, semantic search, and evidence-grounded literature review generation.
- Developed **53 unit and integration tests with pytest** and automated backend deployment using **GitHub Actions** and Hugging Face Spaces.

Smart-FAQ – Semantic Search Platform

GitHub

- Developed a full-stack semantic FAQ platform using **FastAPI, React/Vite, MongoDB, and SentenceTransformers** for natural-language search.
- Implemented embedding-based semantic retrieval, confidence scoring, fallback handling, and JWT-protected administrative REST APIs.

RESEARCH & ACHIEVEMENTS

- Research Presenter – CIS 2026:** Presented Saaram research at the **7th Congress on Intelligent Systems (CIS 2026)**, NIT Warangal.
- Best Project Award**, Malla Reddy University – Recognized for the Telugu NLP news summarization project.

CERTIFICATIONS

- Machine Learning Specialization** – DeepLearning.AI & Stanford University (2026)
- IBM Full Stack Software Developer Professional Certificate** – IBM / Coursera (2024)